

SAI PRANAV MADUPU

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SUMMARY

Motivated and detail-oriented, I have hands-on experience developing AI and software solutions through research and practical implementation. I am skilled in machine learning, data analysis, and simulations, and have strong programming abilities. I enjoy working collaboratively to solve technical problems and deliver effective results.

EXPERIENCE

Tata Consultancy Services - Research

Pune, India

Project Intern - Media & Entertainment Team

May 2025 - July 2025

- Developed an AI-Assisted Music Loop Recommender for DJs and Live Musicians, creating a system to select optimal music loops for live performances by analyzing real-time musical flow and style.
- Applied machine learning and audio signal analysis to enhance loop recommendations, supporting dynamic performance adaptability and ensuring perceptually relevant results.

EDUCATION

IIITDM KANCHEEPURAM

Chennai, India

B.Tech in Computer Science Engineering (CGPA: 8.22/10)

2022 - 2026(Pursuing)

NARAYANA JUNIOR COLLEGE

Hyderabad, India

11th and 12th Standard (Marks: 967/1000)

2020 - 2022

PANINEEYA MAHAVIDYALAYA PUBLIC SCHOOL

Hyderabad, India

10th Standard (Marks: 461/500)

2019 - 2020

PROJECTS

Ingest AI: Document QA Chatbot

Skills: Python, Pytorch, Transformers, Streamlit, Milvus Cloud DB, FastAPI, Docker

- **Developed a scalable Retrieval-Augmented Generation (RAG) system** supporting multi-format document ingestion (.pdf, .docx, .csv, .txt) with semantic search and contextual Q&A across documents.
- **Implemented end-to-end NLP pipeline** using BAAI/llm-embedder for embedding, Milvus Cloud for vector storage, and Phi-3 Mini 128K Instruct (4-bit) for efficient response generation.
- **Built an interactive Streamlit-based UI** enabling users to upload, explore, and query large document sets with fast, memory-efficient, and context-aware responses.

AI Based Intrusion Detection System: Botnet Detection

Skills: Python, FastAPI, Pytorch, Docker

- Designed and trained a 1D-CNN model on network flow features, achieving **88% F1-score**, **91% accuracy**, and **83% recall** for botnet classification across 78k+ traffic samples.
- Containerized the model using **FastAPI and Docker**, enabling **real-time inference** and seamless integration with network traffic collectors for proactive threat detection.

Appointment Scheduler Service

Skills: Node.js, Express.js, REST API, Docker, Azure VM, Postman, OCR

- Designed and deployed a scalable API service that parses appointment details from text or images using **Mistral AI OCR + LLM models**, enabling intelligent scheduling automation.
- Containerized the application with **Docker** and deployed it on an **Azure Virtual Machine**, with testing and validation performed through **Postman** for both text and image-based inputs.

RESEARCH AND PUBLICATIONS

- Roy Chaudhuri, S., **Madupu, S.P.**, Sai, K.T.G., Jamwal, V. (2026). LoopMatcher: Proof-of-Concept for AI-Assisted Music Loop Search. In: Machado, P., Romero, J.J., Rebelo, S.M. (eds) Artificial Intelligence in Music, Sound, Art and Design. EvoMUSART 2026. Lecture Notes in Computer Science, vol 16523. Springer, Cham.
https://doi.org/10.1007/978-3-032-24350-8_26

SKILLS

- Programming Languages:** Python, C, C++, SQL
- Web & Backend Development:** FastAPI, HTML, CSS, JavaScript, MySQL, SQLite, MongoDB, React.js, Next.js, Node.js, Express.js
- Machine Learning & LLMs:** Scikit-learn, TensorFlow, PyTorch, Transformers (Hugging Face), OpenCV, Langchain
- Data Science & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- Tools & Platforms:** Git, Azure, Docker, BeautifulSoup, Selenium, Linux

POSITION OF RESPONSIBILITIES & ACHIEVEMENTS

- Finalist* in AI4Andhra Police Hackathon 2025
- Sponsorship Lead*, Savāra 2026, IIITDM Kancheepuram 2026
- Head Core*, Developers Club, IIITDM Kancheepuram 2025 - 2026
- Hospitality Co-Lead*, Samgatha X Vashisht 2025, IIITDM Kancheepuram 2025